6. **Hospital Management System**

Program :

```
from abc import ABC, abstractmethod


# Abstraction
class Patient(ABC):


    def __init__(self):
        pass


    @abstractmethod
    def calculate_bill(self):
        pass


# Inheritance
class InPatient(Patient):


    def __init__(self):
        super().__init__()
        self.room_charge = 15000
        self.treatment_charge = 5000


    # Polymorphism (Overriding)
    def calculate_bill(self):
        return self.room_charge + self.treatment_charge
```

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```
# ---- Main Program ----

patient_type = input("Enter Patient Type (InPatient): ")

if patient_type.lower() == "inpatient":
    p = InPatient()
else:
    print("Invalid Patient Type")
    exit()

print("Total Bill:", p.calculate_bill())
```

Output :

Enter Patient Type (InPatient): InPatient

Total Bill: 20000

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